Convolution as a local operation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable local-filter evidence record.

Trace windows,verify shapes and distinguish arithmetic responses from learned recognition claims.

Keep one habit: W1–W3/W5 → selected new filter → project operation ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use a fictional single-channel image with rows [1, 2, 0], [0, 1, 3] and [2, 0, 1]. Coordinates are zero-based row then column. Our **kernel** has rows [1, -1] and [0, 1].

  At each position, multiply matching entries of a 2 by 2 window and the **kernel**, then add the four products. Use bias 0, **stride** 1 and no **padding**.

- At the top-left window, the sum is 1 times 1 plus 2 times -1 plus 0 times 0 plus 1 times 1 = 0. Slide the same **kernel** right, then down. Only use windows that fit entirely inside the image; this is the valid boundary rule.

- The **feature map** has rows [0, 5] and [-1, -1]. This neural-network convention keeps the **kernel** orientation fixed; mathematically it is cross-correlation. Mathematical convolution reverses the **kernel** along both spatial axes.

  We use the fixed-orientation convention throughout, so every hand calculation and program agrees.

Image tensors and channels

> **My remembered idea, example and correction**

Debugging shapes and gradients

> **My remembered idea, example and correction**

Training and validation curves

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Window start	Four products	Sum
[0, 0]	1, -2, 0, 1	0
[0, 1]	2, 0, 0, 3	5
[1, 0]	0, -1, 0, 0	-1
[1, 1]	1, -3, 0, 1	-1

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Mainimage[[1, 2, 0], [0, 1, 3], [2, 0, 1]], kernel[[1, –1], [0, 1]], fixedorientation, stride 1, no padding, bias 0. Row-major products[1, –2, 0, 1]→0; [2, 0, 0, 3]→5; [0, –1, 0, 0]→–1; [1, –3, 0, 1]→–1, map[[0, 5], [–1, –1]]. Four shared weights, 16 direct products; one shared bias gives 5 parameter slots. Cornerinput1→4 changes only first output to3. Stride2→[[0]]; onezero border→4×4 output, top-left 1.

  ReLUoriginalmap[[0, 5], [0, 0]]. Flippedkernel[[1, 0], [–1, 1]]top-left 2. Separate pictureimage[[1, 2, 3], [4, 5, 6]], kernel[[1, 0], [0, –1]]→[[-4, –4]], bias 2→[[-2, –2]]. Kernels hand-chosen, not learned recognition.

- Use the lesson image and **kernel** at window start [0, 1]. List the window rows, four products and **output** before activation.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Mainimage[[1, 2, 0], [0, 1, 3], [2, 0, 1]], kernel[[1, –1], [0, 1]], fixedorientation, stride 1, no padding, bias 0. Row-major products[1, –2, 0, 1]→0; [2, 0, 0, 3]→5; [0, –1, 0, 0]→–1; [1, –3, 0, 1]→–1, map[[0, 5], [–1, –1]]. Four shared weights, 16 direct products; one shared bias gives 5 parameter slots. Cornerinput1→4 changes only first output to3. Stride2→[[0]]; onezero border→4×4 output, top-left 1.

  ReLUoriginalmap[[0, 5], [0, 0]]. Flippedkernel[[1, 0], [–1, 1]]top-left 2. Separate pictureimage[[1, 2, 3], [4, 5, 6]], kernel[[1, 0], [0, –1]]→[[-4, –4]], bias 2→[[-2, –2]]. Kernels hand-chosen, not learned recognition.

- How many shared **kernel** weights does the lesson use? How many products are computed over all four valid windows? What changes if one shared bias is added?

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Mainimage[[1, 2, 0], [0, 1, 3], [2, 0, 1]], kernel[[1, –1], [0, 1]], fixedorientation, stride 1, no padding, bias 0. Row-major products[1, –2, 0, 1]→0; [2, 0, 0, 3]→5; [0, –1, 0, 0]→–1; [1, –3, 0, 1]→–1, map[[0, 5], [–1, –1]]. Four shared weights, 16 direct products; one shared bias gives 5 parameter slots. Cornerinput1→4 changes only first output to3. Stride2→[[0]]; onezero border→4×4 output, top-left 1.

  ReLUoriginalmap[[0, 5], [0, 0]]. Flippedkernel[[1, 0], [–1, 1]]top-left 2. Separate pictureimage[[1, 2, 3], [4, 5, 6]], kernel[[1, 0], [0, –1]]→[[-4, –4]], bias 2→[[-2, –2]]. Kernels hand-chosen, not learned recognition.

- For the lesson image and **kernel**, give the **output** shape with no **padding** and **stride** 2. Then give the shape with one zero border on each side and **stride** 1, and calculate that padded map's top-left **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Mainimage[[1, 2, 0], [0, 1, 3], [2, 0, 1]], kernel[[1, –1], [0, 1]], fixedorientation, stride 1, no padding, bias 0. Row-major products[1, –2, 0, 1]→0; [2, 0, 0, 3]→5; [0, –1, 0, 0]→–1; [1, –3, 0, 1]→–1, map[[0, 5], [–1, –1]]. Four shared weights, 16 direct products; one shared bias gives 5 parameter slots. Cornerinput1→4 changes only first output to3. Stride2→[[0]]; onezero border→4×4 output, top-left 1.

  ReLUoriginalmap[[0, 5], [0, 0]]. Flippedkernel[[1, 0], [–1, 1]]top-left 2. Separate pictureimage[[1, 2, 3], [4, 5, 6]], kernel[[1, 0], [0, –1]]→[[-4, –4]], bias 2→[[-2, –2]]. Kernels hand-chosen, not learned recognition.

- A program reverses both axes of the lesson **kernel** to [[1, 0], [-1, 1]]. Compute the top-left **output**. Does it agree with our declared convention? Explain what the test checks.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use image rows [2, 1, 0], [1, 3, 2] and **kernel** rows [0, 1], [-1, 2]. Use no **padding**, **stride** 1 and bias 0. Give the **output** shape, both four-product sums and the **feature map**. Then add a shared bias of -5 and apply ReLU.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Filter a new image independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled convolution as a local operation **evidence** sheet to Image-pattern Network Investigation. Show how you can compute a local weighted sum at each valid image position.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Separate parameters and operations for Convolution as a local operation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Convolution as a local operation. Proposed explanation: "Each window has its own set of learned weights". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use image rows [2, 1, 0], [1, 3, 2] and **kernel** rows [0, 1], [-1, 2]. Use no **padding**, **stride** 1 and bias 0. Give the **output** shape, both four-product sums and the **feature map**. Then add a shared bias of -5 and apply ReLU.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

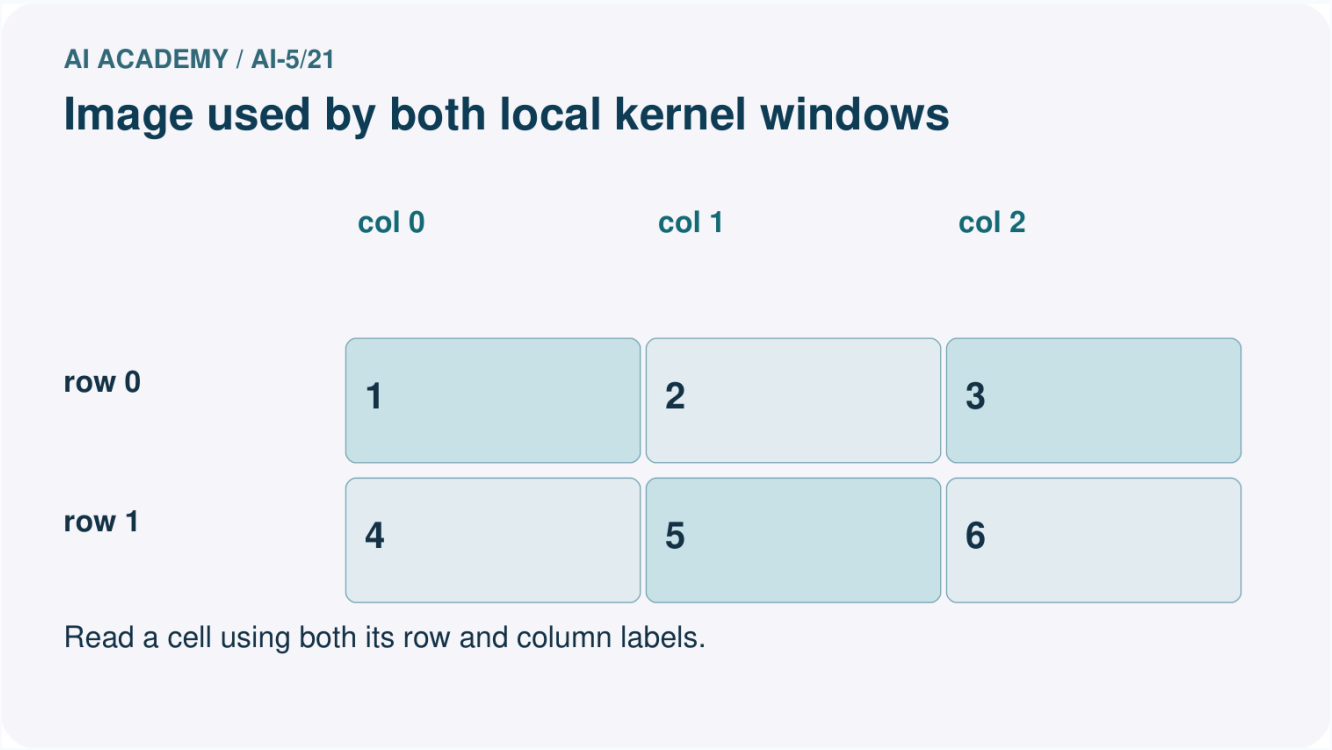
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Draw the exact patch-by-patch arithmetic of a local image operation.
- Image rows [1, 2, 3],[4, 5, 6]; **kernel** rows [1, 0],[0,-1]. Use unflipped **kernel**, no **padding**, **stride** 1, bias 0.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Why is flipping the **kernel** an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/21

Convolution as a local operation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. First patch

1 x 1+2 x 0+4 x 0+5 x (-1)=-4.

↓

2. Second patch

2 x 1+3 x 0+5 x 0+6 x (-1)=-4.

↓

3. Feature map

Output shape 1 x 2 and map [[-4,-4]]; the same kernel is reused at each position.

CHANGE ONE THING

Add shared bias 2 after each patch sum.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Draw the exact patch-by-patch arithmetic of a local image operation.
- Which labelled step matches this detail? **Output** shape 1×2 and map [[-4,-4]]; the same **kernel** is reused at each position.
- 1. **Feature map**
- 2. First patch
- 3. Second patch
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: **Feature map Output** shape 1×2 and map [[-4,-4]]; the same **kernel** is reused at each position.

A local **kernel** combines the values in one small image window with a fixed set of weights. Slide that same **kernel** to the next valid window and repeat the products.

Sharing weights lets the operation look for the same local pattern at different positions. This example declares an unflipped **kernel**; adding bias changes each window result without changing which pixels were used.

Changed-case question and optional hint

- Changed-case practice: Add shared bias 2 after each patch sum. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. **Feature map** becomes [[-2,-2]].
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: **Feature map** becomes [[-2,-2]]. Bias adds 2 to each **output**; it does not change patch products or **output** shape.

Repair and retain the record

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** and validation curves (ai-5/15). Goal: Compute comparable **training** and validation losses at saved checkpoints. Short practice: Can the original epoch 3 **training** loss 1 justify claiming best generalisation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Convolution as a local operation: the case

AI ACADEMY / AI-5/21 · 01 · LOOK AT THE FACTS

Convolution as a local operation: the case

The situation

Draw the exact patch-by-patch arithmetic of a local image operation.

Supplied evidence

Image rows [1, 2, 3],[4, 5, 6]; kernel rows [1, 0],[0,-1]. Use unflipped kernel, no padding, stride 1, bias 0.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain convolution as a local operation using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

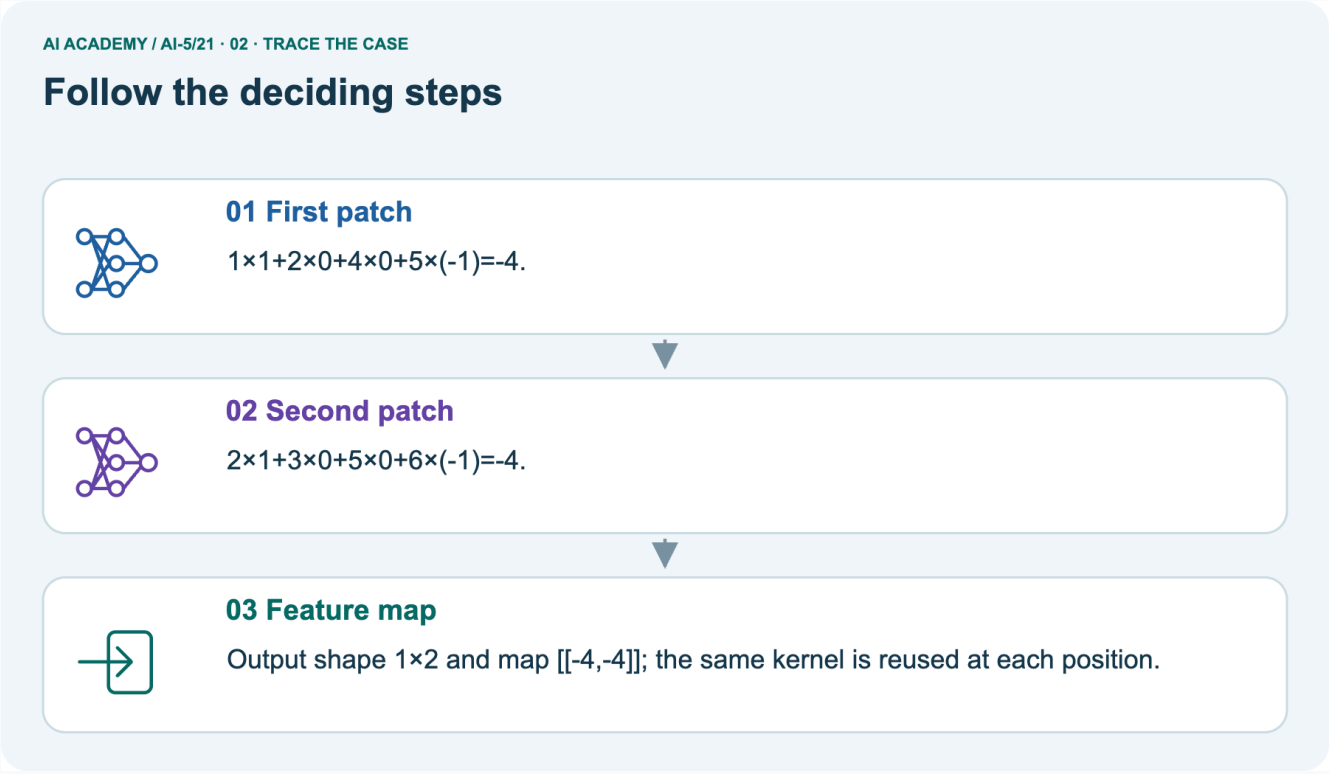
Draw the exact patch-by-patch arithmetic of a local image operation. Supplied facts: Image rows [1, 2, 3],[4, 5, 6]; **kernel** rows [1, 0],[0,-1]. Use unflipped **kernel**, no **padding**, **stride** 1, bias 0.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First patch: 1×1+2×0+4×0+5×(-1)=-4. → Second patch: 2×1+3×0+5×0+6×(-1)=-4. → **Feature map**: **Output** shape 1×2 and map [[-4,-4]]; the same **kernel** is reused at each position.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Image used by both local kernel windows



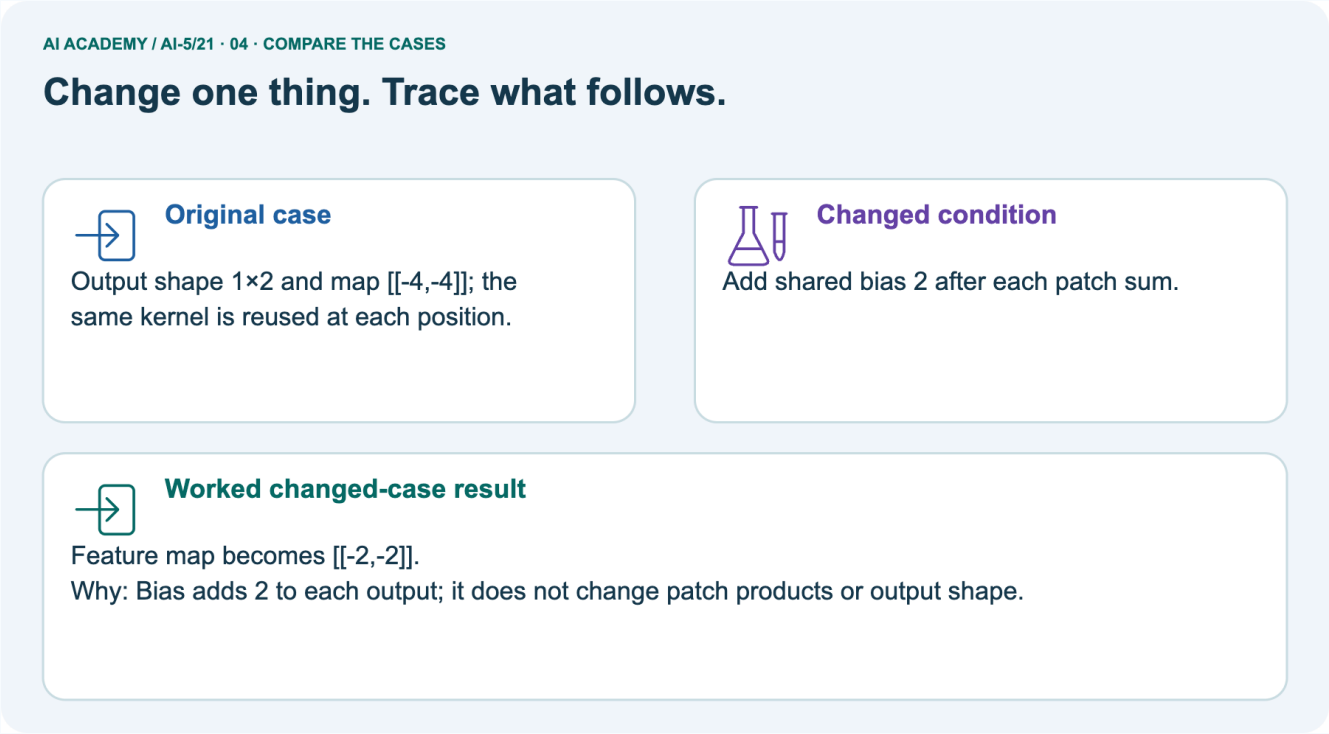
Follow the labelled picture with the matching explanation. Open the vector diagram

Image used by both local **kernel** windows. row 0: col 0 = 1, col 1 = 2, col 2 = 3. row 1: col 0 = 4, col 1 = 5, col 2 = 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

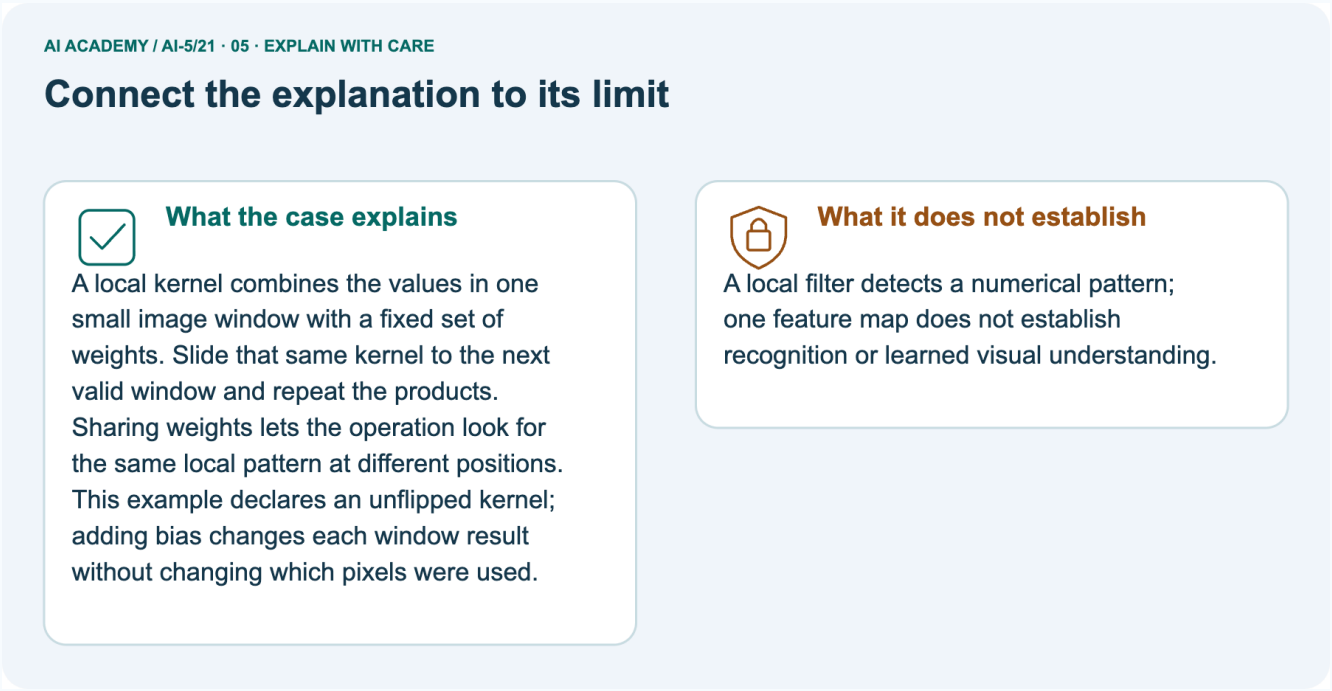
Original case: **Output** shape 1×2 and map [[-4,-4]]; the same **kernel** is reused at each position. Changed condition: Add shared bias 2 after each patch sum. Expected result: **Feature map** becomes [[-2,-2]].

Reason: Bias adds 2 to each **output**; it does not change patch products or **output** shape.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A local **kernel** combines the values in one small image window with a fixed set of weights. Slide that same **kernel** to the next valid window and repeat the products.

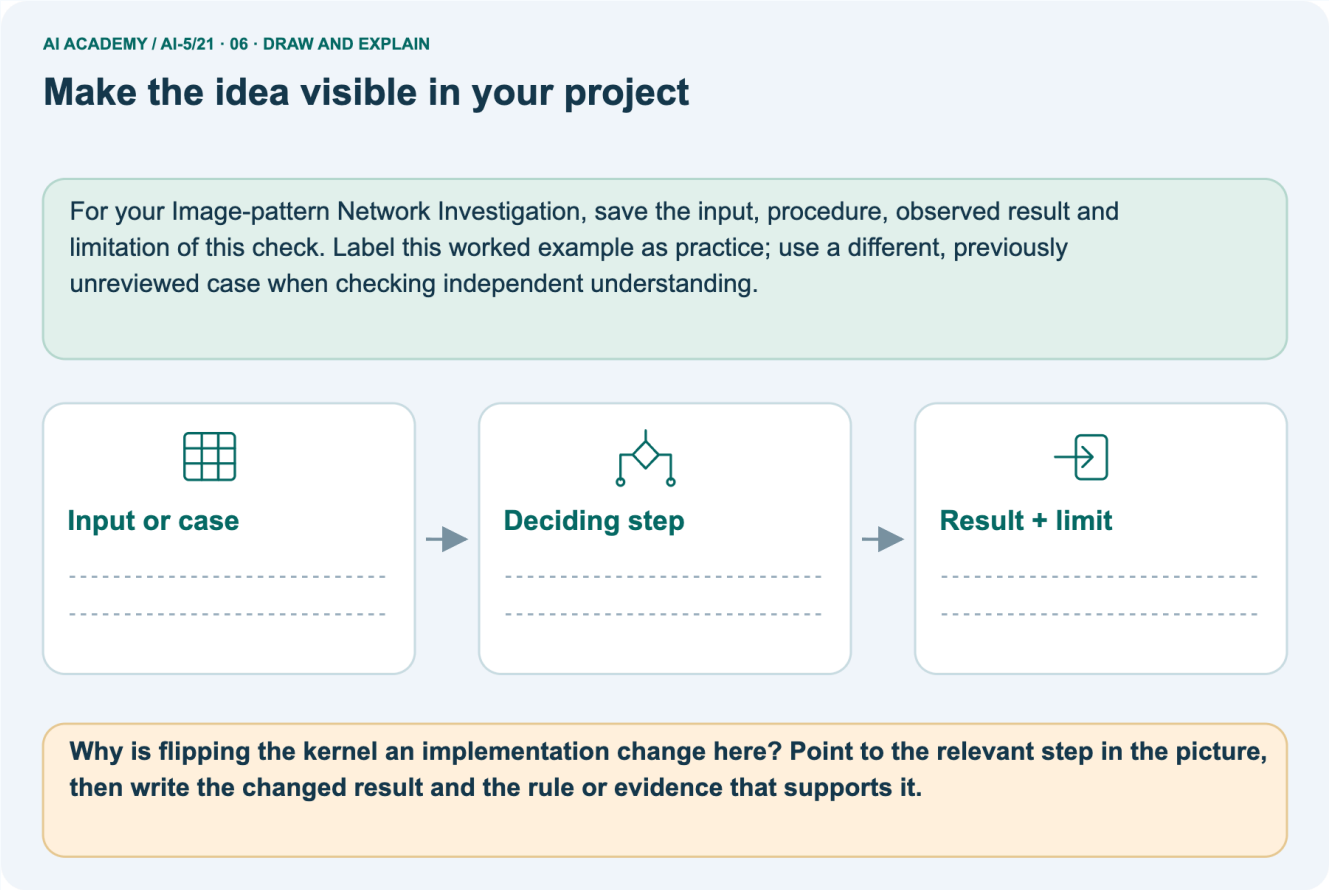
Sharing weights lets the operation look for the same local pattern at different positions. This example declares an unflipped **kernel**; adding bias changes each window result without changing which pixels were used.

Limit: A local filter detects a numerical pattern; one **feature map** does not establish recognition or learned visual understanding.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is flipping the **kernel** an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Convolution as a local operation: the case

02. Follow the deciding steps

03. Image used by both local kernel windows

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.